

Notes: Lang, Ofek, and Stulz (JFE, 1996)

Leverage, Investment, and Firm Growth

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1 Big picture

This paper asks whether leverage affects real corporate growth. The basic empirical fact is that firms with more debt subsequently grow less. The paper's main contribution is to show that this fact is not just a simple correlation: it appears in firm-level investment and employment, in business segments inside diversified firms, and across a variety of leverage and growth measures.

Main finding

The paper's central result is that **leverage is negatively related to future growth, but primarily for low- q firms**. High- q firms, which the market recognizes as having valuable growth opportunities, do not exhibit the same strong negative relation between leverage and growth.

- At the firm level, book leverage predicts lower net investment, lower employment growth, and lower capital-expenditure growth.
- The relation survives industry adjustment and controls for cash flow, past capital expenditures, sales growth, and Tobin's q .
- At the segment level, even noncore segments of diversified firms grow less when the parent firm has higher leverage.
- The leverage-growth relation is concentrated among firms with $q < 1$, not firms with $q > 1$.

Interpretation: leverage matters for real decisions. The evidence is consistent with debt overhang or liquidity constraints for firms whose investment opportunities are poor, marginal, or not fully recognized by outside investors. It is also consistent with the disciplinary role of debt if low- q firms would otherwise invest in poor projects.

2 Incremental contribution of the paper

1. **Direct evidence on leverage and real growth.** The paper moves beyond cash-flow sensitivity and tests whether debt itself predicts investment, employment, and capital-expenditure growth.
2. **Separates leverage from cash flow.** By using cash flow gross of interest and separately examining debt service, the paper shows that a dollar of debt-service pressure is associated with a larger decline in growth than a dollar decline in operating cash flow.
3. **Segment-level test for omitted growth opportunities.** The paper uses noncore business segments of diversified firms to address the concern that leverage merely proxies for unobserved growth opportunities.
4. **Heterogeneity by investment opportunities.** The paper shows that leverage predicts lower growth for low- q firms but not for high- q firms, linking the effect of leverage to the quality or market recognition of investment opportunities.
5. **Robustness across measures and samples.** The paper checks alternative leverage definitions, alternative growth-opportunity measures, better-performing firms, 1980s subsamples, and smaller firms.

3 Research question and hypotheses

The paper is motivated by two competing views of capital structure.

3.1 Irrelevance or self-correction view

If capital markets can fund all valuable projects, leverage should not prevent good firms from growing. Under this view, a negative leverage-growth relation could simply mean that firms with fewer growth opportunities optimally choose more debt.

3.2 Debt overhang and liquidity view

High leverage can reduce growth by lowering internal funds available for investment and by raising the cost of external finance. In the Myers-style debt-overhang logic, shareholders may avoid financing positive-NPV projects if too much of the value accrues to existing debt holders.

3.3 Disciplinary debt view

Debt can also reduce inefficient growth. If managers of low- q firms would otherwise invest free cash flow in poor projects, leverage can increase value by limiting wasteful investment.

4 Data and measurement

4.1 Firm-level sample

The base sample uses Compustat industrial firms over 1970–1989. Firms enter a base-year sample if they have at least \$1 billion of sales in 1989 dollars. The authors restrict attention to industrial firms and require data on sales, employees, and capital expenditures. The sample contains 640 different firms, with 142 firms satisfying the criteria in every year.

4.2 Main variables

- **Investment:** capital expenditures minus depreciation in year +1, scaled by fixed assets at year 0.
- **Capital-expenditure growth:** real growth in capital expenditures from year 0 to year +1, and in some tests to year +3.
- **Employment growth:** growth in the number of employees from year 0 to year +1, and in some tests to year +3.
- **Book leverage:** book value of short-term plus long-term debt divided by book assets.
- **Cash flow:** cash flow gross of interest, scaled by assets. This avoids mechanically absorbing leverage through interest payments.
- **Tobin's q :** market value of equity plus book debt, divided by replacement cost of assets.

4.3 Segment-level sample

To separate leverage from unobserved growth opportunities, the authors use the Compustat segment files from 1986 to 1991. They classify a segment as a core segment if it matches the firm’s main SIC code and classify the remaining segments as noncore. The key idea is that noncore segment opportunities should have less influence on firm-level capital structure, so a leverage effect in noncore segments is harder to explain as leverage merely proxying for growth opportunities.

5 Empirical design

5.1 Firm-level growth regressions

The baseline specification regresses future growth on leverage, controls, and year indicators:

$$Growth_{i,t+1} = \alpha_t + \beta Leverage_{i,t} + \gamma CashFlow_{i,t} + \delta X_{i,t} + \varepsilon_{i,t+1}.$$

The controls include capital expenditures, sales growth, and Tobin’s q . The main coefficient is β . A negative β means that more levered firms grow less after conditioning on standard predictors of growth.

5.2 Industry-adjusted regressions

The authors repeat the tests after subtracting industry medians from the variables. This checks whether the results are simply driven by persistent industry leverage differences.

5.3 Segment regressions

The segment tests regress segment capital expenditures on parent leverage and segment return on assets. If leverage only proxies for firm-level growth opportunities, the coefficient should be much weaker for noncore segments. It is not.

5.4 Low- q versus high- q tests

The authors interact leverage with whether a firm has $q < 1$ or $q > 1$. This tests whether leverage restricts growth mainly when outside investors do not recognize valuable investment opportunities.

6 Tables and figures

6.1 Summary statistics and raw correlations

Table 1
Growth, leverage, and investment opportunity set measures

The sample period is 1970–1989. Included firms have \$1 billion of sales in 1989 dollars for each year in which they appear in the sample. All data are obtained from Compustat. Investment is capital expenditures minus depreciation for year $t + 1$ divided by the book value of fixed assets at the end of year t . Employment growth is the percent change in employment. Capital expenditures growth is the percent change in capital expenditures adjusted for inflation. Book leverage is debt divided by total assets (TA). Market leverage is debt divided by book debt plus the market value of equity. All growth measures are obtained by comparing a variable at the end of the year in which the firm enters the sample to its value at the end of the following (third) year for one-year (three-year) growth measures. Cash flow is gross of interest payments.

	Mean (<i>t</i> -statistic)	25th percentile	Median	75th percentile	Standard deviation (# of firm- years)
Net investment/ fixed assets	0.122 (68.04)	0.041	0.096	0.164	0.148 (6,945)
1-year employment growth	0.013 (6.84)	– 0.046	0.003	0.054	0.163 (7,023)
3-year employment growth	0.036 (9.12)	– 0.114	0.003	0.136	0.294 (5,645)
1-year capital expenditures growth	0.111 (17.48)	– 0.196	0.024	0.284	0.536 (7,040)
3-year capital expenditures growth	0.237 (23.59)	– 0.264	0.074	0.514	0.754 (5,648)
Cash flow/TA	0.106 (152.42)	0.079	0.104	0.133	0.058 (6,915)
Tobin's <i>q</i>	0.961 (88.04)	0.509	0.715	1.083	0.908 (6,929)
Book leverage	0.243 (164.22)	0.164	0.234	0.307	0.124 (7,049)
Market leverage	0.323 (134.28)	0.171	0.291	0.448	0.203 (7,049)

Table 1. Growth, leverage, and investment-opportunity measures.

Read:

- Table 1 shows that the average firm has net investment of about 12.2 percent of fixed assets, book leverage of about 24.3 percent, and Tobin's *q* near 0.96.
- The sample includes several growth measures: net investment, employment growth, and capital-expenditure growth at one-year and three-year horizons.
- Table 2 shows that book leverage is negatively correlated with all growth measures.
- Tobin's *q* is positively related to growth and negatively related to leverage, which motivates the multivariate and low-*q*/high-*q* tests.

Economic message: the raw data already contain a negative leverage-growth relation, but this relation could reflect omitted investment opportunities. The rest of the paper is designed to separate these explanations.

Table 2
Correlations among variables used in this study

The first line gives the correlation between two variables using the raw data; the second line gives the correlation using industry-adjusted variables. Industry-adjusted variables are obtained by subtracting the industry median at the four-digit SIC level. Net investment is capital expenditures minus depreciation in year $t + 1$ divided by fixed assets (FA) at the end of year t . Capital expenditures growth is the percent change in capital expenditures adjusted for inflation. Employment growth is the percent change in employment. Cash flow is gross of interest expense.

	Net investment	1-year capital expend- iture growth	3-year capital expend- iture growth	1-year employ- ment growth	3-year employ- ment growth	Book leverage
Net investment	1.00	0.65*	0.08*	0.38*	0.23*	– 0.14*
	1.00	0.66*	0.09*	0.37*	0.24*	– 0.10*
1-year capital expenditure growth	0.65*	1.00	0.40*	0.39*	0.22*	– 0.11*
	0.66*	1.00	0.38*	0.37*	0.23*	– 0.09*
3-year capital expenditure growth	0.08*	0.40*	1.00	0.20*	0.39*	– 0.09*
	0.09*	0.38*	1.00	0.16*	0.36*	– 0.08*
1-year employment growth	0.38*	0.39*	0.20*	1.00	0.59*	– 0.08*
	0.37*	0.37*	0.16*	1.00	0.57*	– 0.07*
3-year employment growth	0.23*	0.22*	0.39*	0.59*	1.00	– 0.11*
	0.24*	0.23*	0.36*	0.57*	1.00	– 0.10*
Capital expend- itures (0)/FA	0.26*	– 0.14*	– 0.24*	0.07*	0.05*	0.01
	0.22*	– 0.13*	– 0.23*	0.07*	0.04*	0.035
Tobin's <i>q</i>	0.13*	0.05*	0.07*	0.14*	0.18*	– 0.19*
	0.15*	0.06*	0.05*	0.11*	0.13*	– 0.18*
Cash flow/TA	0.23*	0.06*	– 0.01	0.12*	0.14*	– 0.10*
	0.18*	0.04	0.00	0.12*	0.17*	– 0.02*
Market leverage	– 0.21*	– 0.13*	– 0.12*	– 0.16*	– 0.21*	0.75*
	– 0.17*	– 0.11*	– 0.09*	– 0.13*	– 0.18*	0.76*

*Significant at the 0.01 level.

Table 2. Correlations among variables.

6.2 Baseline leverage-growth regressions

Table 3
Regressions of growth measures on leverage

The sample period is 1970–1989. Included firms have \$1 billion of sales in 1989 dollars for each year in which they appear in the sample. All data are obtained from Compustat. Investment is capital expenditures at year 0 minus depreciation divided by fixed assets (FA) at the end of year 0. Capital expenditures growth is the percent change in capital expenditures adjusted for inflation. Employment growth is the percent change in employment. Book leverage is defined as the ratio of total debt to the book value of total assets, and q is defined as the ratio of the market value of equity plus the book value of debt to the replacement cost of the assets. All explanatory variables are computed for the base year; flow variables are normalized using total assets (TA) at the end of the previous year. p -values are in parentheses.

Dependent variable (# of obs.)	Investment/ FA(0) (6,791)	1-year employ- ment growth (6,777)	3-year employ- ment growth (5,478)	1-year capital expenditures growth (6,795)	3-year capital expenditures growth (5,480)
Intercept	0.002 (0.020)	− 0.041 (0.001)	0.054 (0.019)	0.079 (0.055)	0.365 (0.001)
Book leverage/TA	− 0.105 (0.001)	− 0.066 (0.001)	− 0.200 (0.001)	− 0.480 (0.001)	− 0.634 (0.001)
Cash flow/TA	0.324 (0.001)	0.238 (0.001)	0.643 (0.001)	0.378 (0.039)	0.754 (0.011)
Capital expenditures/ FA(−1)	0.105 (0.001)	0.023 (0.012)	− 0.015 (0.571)	− 0.368 (0.012)	− 1.066 (0.001)
Sales growth	0.015 (0.068)	0.029 (0.007)	0.172 (0.001)	0.277 (0.001)	0.409 (0.001)
Tobin's q	0.017 (0.001)	0.017 (0.001)	0.033 (0.001)	0.017 (0.014)	0.030 (0.016)
R -sq	0.148	0.060	0.094	0.087	0.120

The results in Table 3 do not control for industry effects. One approach to controlling for industry effects, cited in the capital structure literature, is to allow for industry indicator variables (see, for instance, Bradley, Jarrell, and Kim, 1984). The problem with this approach is that we are specifically concerned with

Table 3. Regressions of growth measures on leverage.

Read:

- Table 3 shows a negative book-leverage coefficient for every growth measure. For example, the coefficient is -0.105 for investment over fixed assets and -0.480 for one-year capital-expenditure growth.
- Cash flow, sales growth, and q generally predict more growth, as expected.
- Table 4 repeats the exercise using industry-adjusted variables. The leverage coefficients remain negative and significant.
- The industry-adjusted evidence shows that the result is not just an industry composition effect.

Economic message: leverage predicts lower future growth after controlling for cash flow, prior investment, sales growth, Tobin's q , and year effects. The relation also holds within industries.

Table 4
Industry-adjusted regressions of growth measures on leverage

The sample period is 1970–1989. Included firms have \$1 billion of sales in 1989 dollars for each year in which they appear in the sample. All data are obtained from Compustat. Investment is capital expenditures at year 0 minus depreciation divided by fixed assets (FA) at the end of year 0. Capital expenditures growth is the percent change in capital expenditures adjusted for inflation. Employment growth is the percent change in employment. Book leverage is defined as the ratio of total debt to the book value of total assets, and q is the ratio of the market value of equity plus the book value of debt to the replacement cost of the assets. All explanatory variables are computed for the base year; flow variables are normalized using total assets (TA) at the end of the previous year. p -values are in parentheses.

Dependent variable (# of obs.)	Investment/ FA(0) (6,791)	1-year employ- ment growth (6,777)	3-year employ- ment growth (5,478)	1-year capital expenditures growth (6,795)	3-year capital expenditures growth (5,480)
Intercept	0.002 (0.581)	0.003 (0.602)	0.033 (0.021)	0.081 (0.091)	0.158 (0.001)
Book leverage	− 0.089 (0.001)	− 0.057 (0.008)	− 0.155 (0.001)	− 0.428 (0.001)	− 0.416 (0.001)
Cash flow/TA	0.287 (0.001)	0.256 (0.001)	0.800 (0.001)	0.336 (0.049)	0.755 (0.007)
Capital expenditures (0)/FA	0.088 (0.001)	0.014 (0.184)	− 0.096 (0.001)	− 0.283 (0.009)	− 0.813 (0.001)
Sales growth	0.012 (0.177)	0.029 (0.008)	0.217 (0.001)	0.211 (0.001)	0.255 (0.002)
Tobin's q	0.018 (0.001)	0.018 (0.001)	0.031 (0.001)	0.027 (0.002)	0.045 (0.004)
R -sq	0.079	0.026	0.049	0.036	0.055

Table 4 provides the regression estimates obtained after adjusting all vari-

Table 4. Industry-adjusted regressions of growth on leverage.

6.3 Debt service and segment-level evidence

Table 5
Industry-adjusted regressions of growth measures on debt service

The sample period is 1970–1989. Included firms have \$1 billion of sales in 1989 dollars for the year in which they appear in the sample. All data are obtained from Compustat. Investment is capital expenditures at year 0 minus depreciation divided by fixed assets (FA) at the end of year 0. Capital expenditures growth is the percent change in capital expenditures adjusted for inflation. Employment growth is the percent change in employment. Debt service is defined as interest paid divided by total assets, and q is the ratio of the market value of equity plus the book value of debt to the replacement cost of the assets. All explanatory variables are computed for the base year; flow variables are normalized using total assets (TA) at the end of the previous year. p -values are in parentheses. The last row provides p -values for the test that the coefficient on debt service is equal to minus one times the coefficient on cash flow.

Dependent variable (# of obs.)	Investment/ FA(0) (6,791)	1-year employ- ment growth (6,777)	3-year employ- ment growth (5,478)	1-year capital expenditures growth (6,795)	3-year capital expenditures growth (5,392)
Intercept	0.003 (0.529)	0.003 (0.526)	0.034 (0.016)	0.082 (0.001)	0.159 (0.001)
Debt service	− 0.553 (0.001)	− 0.740 (0.001)	− 2.267 (0.001)	− 2.094 (0.001)	− 1.852 (0.040)
Cash flow/TA	0.292 (0.001)	0.251 (0.001)	0.777 (0.001)	0.368 (0.028)	0.784 (0.006)
Capital expenditures (0)/FA	0.086 (0.001)	0.013 (0.250)	− 0.102 (0.001)	− 0.291 (0.008)	− 0.824 (0.001)
Sales growth	0.011 (0.210)	0.030 (0.007)	0.220 (0.001)	0.202 (0.001)	0.246 (0.003)
Tobin's q	0.019 (0.001)	0.018 (0.001)	0.029 (0.001)	0.033 (0.001)	0.051 (0.001)
R -sq	0.077	0.027	0.055	0.030	0.052
p -value for	0.126	0.010	0.001	0.002	0.279

Table 5. Industry-adjusted regressions using debt service.

Read:

- Table 5 shows that debt service is strongly negatively related to growth. The debt-service coefficient is larger in absolute value than the cash-flow coefficient in most columns.
- This means interest burden is not just another cash-flow component; it appears to have a stronger association with future growth than operating cash flow.
- Table 6 moves inside diversified firms. Both noncore and core segments invest less when parent leverage is higher.
- The noncore segment result is especially important because noncore opportunities should be less central to the firm's initial leverage choice.

Economic message: leverage matters beyond standard cash-flow sensitivity. The segment evidence weakens the alternative interpretation that leverage merely proxies for unobserved growth opportunities.

Table 6
Regressions of investment on leverage using segment data

These regressions divide segments of firms in the Compustat Segment Files from 1986 to 1991 between core business segments (same four-digit SIC code as the firm's primary SIC code) and noncore segments. Only segments with \$20 million in sales of firms that have segments with different one-digit SIC codes are included in a given year. ROA is the return on assets corresponding to the unit of measurement of the dependent variable and is observed in the same year as the dependent variable. Leverage is the firm's book leverage observed the year before. The firm sample includes all industrial firms with sales of \$20 million in a given year. The sample includes 2,566 noncore segment years, 950 core segment years, and 8,707 firm years. p -values are in parentheses.

	Noncore segments	Core segments	Firm
<i>Panel A. Medians of selected variables</i>			
Segment assets/TA	0.188	0.423	1.600
Segment capital expenditures/ Total capital expenditures	0.198	0.172	1.000
<i>Panel B. Capital expenditures minus depreciation to total assets</i>			
Constant	0.025 (0.001)	0.031 (0.001)	0.017 (0.001)
ROA	0.027 (0.040)	0.042 (0.001)	0.151 (0.001)
Leverage	− 0.031 (0.001)	− 0.047 (0.001)	− 0.021 (0.001)
R -sq	0.017	0.042	0.136
<i>Panel C. Capital expenditures to total assets</i>			
Constant	0.076 (0.001)	0.077 (0.001)	0.074 (0.001)
ROA	0.058 (0.001)	0.031 (0.001)	0.031 (0.001)
Leverage	− 0.039 (0.001)	− 0.030 (0.001)	− 0.027 (0.001)
R -sq	0.035	0.018	0.017

Table 6. Investment regressions using segment data.

6.4 Alternative segment definitions and segment leverage measures

Table 7

Regressions of investment on leverage using segment data and different definitions of core segments

These regressions divide segments of firms in the Compustat Segment Files from 1986 to 1991 between core business segments and noncore segments. Only segments with \$20 million in sales of firms that have segments with different one-digit SIC codes are included in a given year. ROA is the return on assets corresponding to the unit of measurement of the dependent variable and is observed in the same year as the dependent variable. CapEx denotes capital expenditures. Leverage is the firm's book leverage observed the year before. TA denotes total assets. *p*-values are in parentheses.

Panel A. Medians of selected variables

	Segment assets/Firm assets		Segment CapEx/Firm CapEx	
	Core	Noncore	Core	Noncore
Panel B definition	0.335	0.165	0.380	0.174
Panel C definition	0.460	0.154	0.528	0.162

Panel B. Core segment defined as any segment that has the same two-digit SIC code as the firm

	CapEx – Depreciation/TA		CapEx/TA	
	Noncore	Core	Noncore	Core
Intercept	0.021 (0.001)	0.034 (0.001)	0.078 (0.001)	0.075 (0.001)
Leverage	– 0.026 (0.001)	– 0.046 (0.001)	– 0.041 (0.001)	– 0.032 (0.001)
ROA	0.051 (0.001)	0.010 (0.539)	0.046 (0.001)	0.056 (0.014)
R-sq	0.025	0.028	0.024	0.033

Panel C. Core segment defined as segment with largest sales

	Noncore		Core	
	Noncore	Core	Noncore	Core
Intercept	0.024 (0.001)	0.034 (0.001)	0.078 (0.001)	0.072 (0.001)
Leverage	– 0.036 (0.001)	– 0.037 (0.001)	– 0.043 (0.001)	– 0.027 (0.001)
ROA	0.039 (0.001)	0.010 (0.628)	0.039 (0.001)	0.077 (0.021)
R-sq	0.026	0.018	0.029	0.036

Table 7. Segment evidence with alternative core-segment definitions.

Read:

- Table 7 shows that the segment-level result is robust to defining core segments by two-digit SIC codes or by largest sales segment.
- Noncore segments still display a negative relation between investment and parent leverage.
- Table 8 shows that the segment result is not specific to book leverage. Interest/assets, market leverage, and long-term leverage generally produce negative coefficients.
- The effect is not weaker for noncore segments in a way that would support the pure proxy story.

Economic message: the noncore-segment evidence is robust. This is one of the paper's strongest identification-style arguments: leverage appears to constrain internal capital allocation, not merely proxy for firm-level investment opportunities.

Table 8

Regressions of investment on leverage using segment data and different definitions of leverage

These regressions divide segments of firms in the Compustat Segment Files from 1986 to 1991 between core business segments (same four-digit SIC code as the firm's primary SIC code) and noncore segments. Only segments with \$20 million in sales of firms that have segments with different one-digit SIC codes are included in a given year. ROA is the return on assets corresponding to the unit of measurement of the dependent variable and is observed in the same year as the dependent variable. The leverage measure is obtained at the firm level the year before. *p*-values are in parentheses.

Panel A. (Capital expenditures – Depreciation)/Segment's assets

	Noncore		Core		Noncore	
	Noncore	Core	Noncore	Core	Noncore	Core
Intercept	0.025 (0.001)	0.030 (0.001)	0.029 (0.001)	0.031 (0.001)	0.023 (0.001)	0.029 (0.001)
Interest/Assets	– 0.292 (0.001)	– 0.379 (0.001)				
Market leverage			– 0.033 (0.001)	– 0.035 (0.001)		
Long-term leverage					– 0.030 (0.001)	– 0.049 (0.006)
ROA	0.026 (0.044)	0.044 (0.001)	0.018 (0.159)	0.035 (0.011)	0.029 (0.023)	0.041 (0.002)
R-sq	0.020	0.046	0.038	0.023	0.022	0.034

Panel B. Capital expenditures/Segment's assets

	Noncore		Core		Noncore	
	Noncore	Core	Noncore	Core	Noncore	Core
Intercept	0.076 (0.001)	0.077 (0.001)	0.080 (0.001)	0.081 (0.001)	0.072 (0.001)	0.074 (0.001)
Interest/Assets	– 0.354 (0.001)	– 0.283 (0.001)				
Market leverage			– 0.040 (0.001)	– 0.034 (0.001)		
Long-term leverage					– 0.035 (0.001)	– 0.025 (0.012)
ROA	0.056 (0.001)	0.032 (0.046)	0.045 (0.006)	0.022 (0.181)	0.060 (0.001)	0.031 (0.053)

Table 8. Segment evidence with alternative leverage definitions.

6.5 Investment-opportunity heterogeneity

Table 9
Investment opportunities and the relation between growth and leverage

The sample period is 1970-1989. Included firms have \$1 billion of sales in 1989 dollars for each year in which they appear in the sample. All data are obtained from Compustat. Investment is capital expenditures in year $t+1$ minus depreciation divided by fixed assets (FA) at the end of year t . Capital expenditures growth is the percent change in capital expenditures adjusted for inflation. Employment growth is the percent change in employment. Book leverage is defined as the ratio of total debt to the book value of total assets, and q is the ratio of the market value of equity plus the book value of debt to the replacement cost of the assets. All explanatory variables are computed for the base year; flow variables are normalized using total assets (TA) at the end of the previous year. p -values are in parentheses.

Dependent variable (# of obs.)	Investment/ FA(0) (6,791)	1-year employ- ment growth (6,777)	3-year employ- ment growth (5,478)	1-year capital expenditures growth (6,795)	3-year capital expenditures growth (5,480)
Intercept	0.025 (0.012)	-0.039 (0.001)	0.055 (0.014)	0.083 (0.048)	0.368 (0.001)
Book leverage, $q < 1$	-0.125 (0.001)	-0.091 (0.001)	-0.249 (0.001)	-0.531 (0.001)	-0.747 (0.001)
Book leverage, $q > 1$	-0.035 (0.175)	0.025 (0.377)	0.011 (0.851)	-0.305 (0.001)	-0.140 (0.282)
Cash flow/TA	0.311 (0.001)	0.222 (0.001)	0.590 (0.001)	0.349 (0.054)	0.632 (0.036)
Capital expend- itures (0)/FA	0.105 (0.001)	0.022 (0.028)	-0.023 (0.407)	-0.370 (0.013)	-1.084 (0.001)
Sales growth	0.013 (0.144)	0.026 (0.011)	0.161 (0.001)	0.271 (0.001)	0.584 (0.001)
Tobin's q	0.012 (0.001)	0.011 (0.001)	0.022 (0.001)	0.007 (0.311)	0.006 (0.634)
R-sq	0.151	0.064	0.101	0.088	0.126

firms are firms with $q > 1$. The p -values for the difference in the leverage

Table 9. Investment opportunities and leverage-growth relation.

Read:

- Table 9 separates the leverage coefficient for firms with $q < 1$ and $q > 1$.
- For low- q firms, leverage is strongly negative in every growth regression. For high- q firms, the coefficients are generally small and often insignificant.
- Table 10 shows that the same pattern survives industry adjustment.
- The difference between the low- q and high- q leverage coefficients is statistically meaningful in the industry-adjusted regressions.

Economic message: the paper's most important heterogeneity result is that leverage does not reduce growth for firms with recognized valuable investment opportunities. The negative effect is concentrated among firms with poor or unrecognized opportunities.

Table 10
Industry-adjusted results for the relation between growth and leverage

The sample period is 1970-1989. Included firms have \$1 billion of sales in 1989 dollars for each year in which they appear in the sample. All data are obtained from Compustat. Investment is capital expenditures at year t minus depreciation divided by fixed assets (FA) at the end of year t . Capital expenditures growth is the percent change in capital expenditures adjusted for inflation. Employment growth is the percent change in employment. Book leverage is the ratio of total debt to the book value of total assets. q is the ratio of the market value of equity plus the book value of debt to the replacement cost of the assets. All explanatory variables are computed for the base year; flow variables are normalized using total assets (TA) at the end of the previous year. All variables are industry-adjusted except that the firm's q is used to compute the interaction variable multiplying book leverage. p -values are in parentheses.

Dependent variable (# of obs.)	Investment/ FA(0) (6,791)	1-year employ- ment growth (6,777)	3-year employ- ment growth (5,478)	1-year capital expenditures growth (6,795)	3-year capital expenditures growth (5,480)
Intercept	0.003 (0.477)	0.004 (0.493)	0.035 (0.015)	0.083 (0.001)	0.162 (0.001)
Book leverage, $q < 1$	-0.118 (0.001)	-0.098 (0.001)	-0.238 (0.001)	-0.516 (0.001)	-0.585 (0.001)
Book leverage, $q > 1$	-0.016 (0.660)	0.047 (0.292)	0.064 (0.412)	-0.205 (0.056)	0.035 (0.846)
Cash flow/TA	0.284 (0.001)	0.252 (0.001)	0.774 (0.001)	0.327 (0.052)	0.700 (0.014)
Capital expend- itures (0)/FA	0.088 (0.001)	0.015 (0.182)	-0.095 (0.001)	-0.282 (0.001)	-0.812 (0.001)
Sales growth	0.011 (0.259)	0.027 (0.011)	0.213 (0.001)	0.206 (0.001)	0.247 (0.003)
Tobin's q	0.020 (0.001)	0.020 (0.001)	0.036 (0.001)	0.032 (0.001)	0.055 (0.002)
R-sq	0.086	0.027	0.052	0.037	0.056
p -value for difference in leverage coefficients	0.011	0.004	0.001	0.016	0.004

Table 10. Industry-adjusted low- q versus high- q results.

6.6 Alternative leverage and investment-opportunity measures

Table 11

Alternative leverage measures

The sample period is 1970–1989. Included firms have \$1 billion of sales in 1989 dollars for each year in which they appear in the sample. All data are obtained from Compustat. Investment is capital expenditures at year 0 minus depreciation divided by fixed assets (FA) at the end of year 0. Employee growth is the percent change in employment. Book leverage is the ratio of total debt to the book value of total assets at the end of the base year (T). Market leverage is debt divided by the sum of debt plus the market value of equity. Interest denotes interest expense. Equity is the book value of equity. p -values are in parentheses.

Dependent variable (# of obs.)	Investment/ FA(0) (6,791)	1-year employment growth (6,777)	Investment/ FA(0) (6,791)	1-year employment growth (6,777)	Investment/ FA(0) (6,780)	1-year employment growth (6,766)	Investment/ FA(0) (6,791)	1-year employment growth (6,777)
Intercept	0.005 (0.244)	0.006 (0.255)	0.003 (0.491)	0.004 (0.485)	0.003 (0.495)	0.003 (0.508)	0.003 (0.464)	0.003 (0.575)
Market leverage, $q < 1$	– 0.097 (0.001)	– 0.085 (0.001)						
Market leverage, $q > 1$	– 0.089 (0.001)	– 0.033 (0.294)						
Interest/TA, $q < 1$			– 0.683 (0.001)	– 0.953 (0.001)				
Interest/TA, $q > 1$			– 0.015 (0.969)	0.149 (0.682)				

Table 11. Alternative leverage measures, part 1.

Book equity/TA $q < 1$					0.052 (0.001)	0.054 (0.001)		
Book equity/TA $q > 1$					– 0.008 (0.749)	– 0.037 (0.253)		
Long-term debt/TA, $q < 1$							– 0.125 (0.001)	– 0.086 (0.001)
Long-term debt/TA, $q > 1$							– 0.031 (0.396)	0.034 (0.234)
Cash flow/TA	0.239 (0.001)	0.220 (0.001)	0.293 (0.001)	0.251 (0.001)	0.285 (0.001)	0.243 (0.001)	0.293 (0.001)	0.259 (0.001)
Capital expenditures/ FA	0.089 (0.001)	0.015 (0.167)	0.067 (0.001)	0.013 (0.239)	0.068 (0.001)	0.014 (0.189)	0.068 (0.001)	0.014 (0.186)
Sales growth	0.015 (0.091)	0.030 (0.004)	0.009 (0.323)	0.027 (0.011)	0.009 (0.322)	0.029 (0.006)	0.010 (0.278)	0.026 (0.013)
Tobin's q	0.013 (0.601)	0.015 (0.801)	0.020 (0.001)	0.019 (0.001)	0.020 (0.001)	0.020 (0.001)	0.019 (0.001)	0.020 (0.001)
R-sq	0.087	0.030	0.078	0.028	0.077	0.026	0.080	0.026

Table 11. Alternative leverage measures, part 2.

Read:

- Table 11 checks whether the low- q result depends on the leverage measure.
- Interest/assets, book equity/assets, and long-term debt/assets broadly confirm the finding: leverage-related pressure matters mainly for low- q firms.
- Market leverage behaves differently because it mechanically embeds the market value of equity and therefore can mix leverage with market expectations about growth.

Economic message: the paper's main conclusion is robust to several book-value leverage definitions, but market leverage is less clean because it reflects current market valuation.

Table 12
Alternative measures of growth opportunities

The sample period is 1970–1989. Included firms have \$1 billion of sales in 1989 dollars for each year in which they appear in the sample. All data are obtained from Compustat. Investment is capital expenditures at year 0 minus depreciation divided by fixed assets (FA) at the end of year 0. Employment growth is the percent change in employment. Book leverage is the ratio of total debt to the book value of total assets at the end of the base year (TA). The industry q is median q of the industry. Market-book is the ratio of debt plus the market value of equity and of the book value of assets. Sales growth is the one-year sales growth. p -values are in parentheses.

Dependent variable (# of obs.)	Investment: FA(0) (6,791)	1-year employment growth (6,777)	Investment: FA(0) (6,872)	1-year employment growth (6,858)	Investment: FA(0) (6,791)	1-year employment growth (6,777)
Intercept	0.002 (0.559)	0.003 (0.581)	0.004 (0.313)	0.005 (0.299)	0.002 (0.581)	0.003 (0.611)
Book leverage industry: $q < 1$	–0.103 (0.001)	–0.085 (0.001)				
Book leverage industry: $q > 1$	–0.044 (0.210)	0.031 (0.533)				
Book leverage market-book < 1			–0.146 (0.001)	–0.127 (0.001)		
Book leverage market-book > 1			–0.001 (0.977)	0.005 (0.897)		
Book leverage, sales growth < 0.05				–0.091 (0.001)	–0.081 (0.002)	
Book leverage, sales growth > 0.05				–0.086 (0.002)	–0.024 (0.504)	
Cash flow/TA – 1	0.286 (0.001)	0.253 (0.001)	0.269 (0.001)	0.249 (0.001)	0.287 (0.001)	0.256 (0.001)
Capital expenditures (0)/FA	0.088 (0.001)	0.015 (0.166)	0.088 (0.001)	0.016 (0.131)	0.088 (0.001)	0.014 (0.189)
Sales growth	0.011 (0.225)	0.027 (0.017)	0.016 (0.065)	0.034 (0.003)	0.012 (0.192)	0.027 (0.014)
Tobin's q	0.019 (0.001)	0.020 (0.001)	0.025 (0.001)	0.018 (0.001)	0.018 (0.001)	0.018 (0.001)
R-sq	0.080	0.027	0.080	0.025	0.079	0.026

that leverage is killing off growth opportunities altogether, so that whenever

Table 12. Alternative measures of growth opportunities.

Read:

- Table 12 checks other ways to classify investment opportunities: industry q , market-to-book, and sales growth.
- The low-opportunity/high-opportunity pattern survives with industry q and market-to-book ratios, but not with sales growth.
- Table 13 shows that the finding is not driven only by distressed firms, the 1970s, or very large firms.
- The leverage effect for low- q firms remains in better-performing firms, in the 1980s subsample, and in a smaller-firm sample.

Economic message: the results are strongest when growth opportunities are measured by market-based forward-looking variables. This supports the interpretation that market-recognized investment opportunities shield firms from the negative leverage-growth relation.

Table 13
Alternative samples

The sample period is 1970–1989. The base sample includes firms that have \$1 billion of sales in 1989 dollars for each year in which they appear in the sample. All data are obtained from Compustat. Investment is capital expenditures at year 0 minus depreciation divided by fixed assets (FA) at the end of year 0. Employment growth is the percent change in employment. Book leverage is the ratio of total debt to the book value of total assets at the end of the base year (TA). The industry q is the median q of the industry. Sales growth is the one-year sales growth. ROA denotes return on assets; industry ROA is the industry median. The small-firm sample includes a random sample of firms with more than \$20 million but less than \$1 billion of sales for the year in which they appear in the sample. p -values are in parentheses.

Dependent variable (# of obs.)	Base sample					
	ROA > 0.1767		Decade = 1980s		Small-firm sample	
	Investment: FA(0)	1-year employment growth	Investment: FA(0)	1-year employment growth	Investment: FA(0)	1-year employment growth
Intercept	0.006 (0.281)	0.002 (0.700)	0.007 (0.315)	0.005 (0.551)	0.007 (0.794)	0.002 (0.916)
Book leverage, $q < 1$	–0.116 (0.001)	–0.091 (0.005)	–0.117 (0.001)	–0.112 (0.001)	–0.134 (0.001)	–0.139 (0.001)
Book leverage, $q > 1$	0.000 (0.997)	0.118 (0.271)	0.015 (0.779)	0.073 (0.285)	0.059 (0.109)	–0.019 (0.416)
Cash flow/TA	0.157 (0.021)	0.124 (0.032)	0.202 (0.008)	0.192 (0.002)	0.472 (0.001)	0.346 (0.001)
Capital expenditures (0)/FA	0.124 (0.002)	–0.003 (0.850)	0.084 (0.001)	0.021 (0.075)	0.004 (0.028)	0.001 (0.672)
Sales growth	0.010 (0.326)	0.012 (0.302)	0.002 (0.854)	0.013 (0.200)	0.003 (0.363)	0.002 (0.295)
Tobin's q	0.035 (0.001)	0.037 (0.001)	0.041 (0.001)	0.040 (0.001)	0.035 (0.001)	0.020 (0.001)

Table 13. Alternative samples.

6.7 Year-by-year robustness

base sample altogether. These columns provide estimates for firms too small to belong in the base sample. Since there are many firms with sales of less than \$1 billion, we had to restrict the sample and did so as follows. We select randomly each year the same number of industrial firms with sales between \$20 million and \$1 billion. A firm that appears in the sample in a given year can reappear four years later if it still exists and still satisfies our size restrictions. Clearly, the

Table 14
Summary results for yearly regressions

The regressions of Table 6 are estimated for each year separately. The mean and median coefficients for the leverage variables are reported with the cross-sectional p -values reported in parentheses. The sample period is 1970–1989. Included firms have \$1 billion of sales in 1989 dollars for each year in which they appear in the sample. All data are obtained from Compustat. Investment is capital expenditures at year 0 minus depreciation divided by fixed assets (FA) at the end of year 0. Capital expenditures growth is the percent change in capital expenditures adjusted for inflation. Employment growth is the percent change in employment. Book leverage is the ratio of total debt to the book value of total assets, and q is the ratio of the market value of equity plus the book value of debt to the replacement cost of the assets. BLLQ (BLHQ) is book leverage times an indicator variable that takes the value one for firms with q smaller (greater) than one. p -values are in parentheses.

Dependent variable (# regressions)	Investment/ FA(0) (20)	1-year employ- ment growth (20)	3-year employ- ment growth (17)	1-year capital expenditures growth (20)	3-year capital expenditures growth (17)
Mean BLLQ coefficient (p -value)	− 0.119 (0.001)	− 0.089 (0.001)	− 0.223 (0.001)	− 0.584 (0.001)	− 0.636 (0.001)
Med. BLLQ (p -value) (% positive)	− 0.113 (0.001) 5%	− 0.110 (0.003) 15%	− 0.184 (0.001) 12%	− 0.543 (0.001) 0%	− 0.680 (0.001) 6%
Mean BLHQ coefficient (p -value)	− 0.011 (0.790)	0.042 (0.230)	0.136 (0.011)	− 0.243 (0.048)	0.071 (0.660)
Med. BLHQ (p -value) (% positive)	− 0.013 (0.668) 45%	0.008 (0.323) 55%	0.122 (0.005) 04%	− 0.223 (0.058) 20%	− 0.130 (0.530) 15%

Table 14. Summary results for yearly regressions.

Read:

- Table 14 estimates the key regressions year by year and summarizes the leverage coefficients.
- Low- q leverage coefficients are consistently negative across years.
- High- q leverage coefficients are much closer to zero and are often positive or insignificant.
- This reduces the concern that the main finding is driven by a small number of unusual years.

Economic message: the low- q leverage effect is persistent through time. The paper’s conclusion is not a one-period artifact.

7 Key mechanisms and interpretation

7.1 Why leverage may reduce growth

Leverage can reduce growth through at least two channels. First, interest and debt-service obligations reduce liquidity available for investment. Second, leverage can create debt overhang: when existing debt captures part of the payoff to new projects, shareholders may underinvest even in positive-NPV projects.

7.2 Why the low- q result matters

If leverage were merely a proxy for firms having poor growth opportunities, high- q and low- q splits would not necessarily be so informative. The paper finds that leverage matters mainly when $q < 1$. This suggests that leverage is most relevant where investment opportunities are poor, marginal, or not recognized by outside investors.

7.3 Debt overhang versus discipline

The welfare interpretation is ambiguous. If low- q firms truly have poor projects, leverage may be beneficial because it prevents wasteful growth. If low- q firms actually have good projects that the market fails to recognize, leverage may be costly because it blocks valuable investment.

8 Limitations

- **Causality is difficult.** Leverage is chosen by firms, so it can reflect private information about future growth opportunities. The segment tests help, but they do not fully solve endogeneity.
- **Tobin's q is imperfect.** The low- q versus high- q split relies on market values and replacement-cost measures that may contain measurement error.
- **Sample selection.** The base sample emphasizes large industrial firms with available Compustat data. Results may not fully generalize to private firms, financial firms, utilities, or very small firms.
- **Segment data are limited.** Segment regressions have fewer controls than firm-level regressions, because segment-level data are less complete.
- **Market leverage behaves differently.** Results using market leverage are harder to interpret because market leverage incorporates market expectations about growth.

9 Short summary

Lang, Ofek, and Stulz study whether leverage predicts lower investment and firm growth. Using a 1970–1989 panel of large industrial firms, they find that leverage is negatively related to future investment, employment growth, and capital-expenditure growth. This relation is robust to industry adjustment and controls for cash flow, sales growth, and Tobin's q . Segment-level evidence shows that noncore segments of diversified firms also grow less when the parent has higher leverage, which weakens the explanation that leverage only proxies for unobserved growth opportunities. The key heterogeneity result is that leverage predicts lower growth for low- q firms but not for high- q firms.

10 Conclusion

The paper's main lesson is that leverage can matter for real corporate growth, but its effect is not uniform. Debt does not appear to constrain firms whose valuable investment opportunities are recognized by the market. Instead, the negative leverage-growth relation is concentrated among low- q firms. This can be interpreted either as debt overhang for firms with unrecognized opportunities

or as disciplinary debt preventing overinvestment by firms with poor opportunities. Either way, the paper shows that capital structure is closely connected to investment and growth.